

EC601 — Control System

Semester VI • 3L:0T:0P • 3 Credits

Module 2 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Stability of Feedback Control Systems

Stability is the most fundamental requirement of any control system. An unstable system produces unbounded output even for a bounded input, making it practically useless.

BIBO Stability: A system is **Bounded-Input Bounded-Output (BIBO) stable** if every bounded input produces a bounded output. For an LTI system, this requires all poles of the closed-loop transfer function to lie in the **left half of the s -plane** (i.e., have negative real parts).

1.1 Stability Conditions

Condition	Meaning
Stable	All closed-loop poles have negative real parts (left half s -plane). Output decays to zero.
Marginally Stable	Poles on the imaginary axis (zero real part). Output oscillates with constant amplitude.
Unstable	At least one pole in the right half s-plane . Output grows without bound.

1.2 Relative Stability

Absolute stability only tells us *whether* a system is stable. **Relative stability** tells us *how stable* — i.e., how far the closed-loop poles are from the imaginary axis.

Relative Stability: A measure of the degree of stability. A system with poles far into the left half-plane is *more stable* (faster decay, less oscillation) than one with poles close to the imaginary axis. Quantified by **gain margin** and **phase margin** (covered in Module 3).

Key Concept

If all poles of $T(s) = \frac{G(s)}{1 + G(s)H(s)}$ satisfy $\text{Re}(s_i) < 0$, the system is stable.

The **characteristic equation** $1 + G(s)H(s) = 0$ determines the pole locations.

2. Routh–Hurwitz Stability Criterion

The Routh–Hurwitz criterion determines stability **without solving** the characteristic equation. It counts the number of roots in the right half s -plane using a tabular method.

Definition: The **Routh–Hurwitz criterion** states that a system is stable if and only if all elements of the **first column** of the Routh array are **positive**. The number of sign

changes in the first column equals the number of closed-loop poles in the right half s -plane.

2.1 Constructing the Routh Array

Given the characteristic equation:

$$a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$$

Step 1 — Necessary condition: All coefficients a_i must be present and have the **same sign**. If any coefficient is zero or negative, the system is *unstable or marginally stable* (no need to proceed).

Step 2 — Form the array:

Routh Array Structure

s^n	a_n	a_{n-2}	a_{n-4}	$\dots$
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	$\dots$
s^{n-2}	b_1	b_2	b_3	$\dots$
s^{n-3}	c_1	c_2	c_3	$\dots$
$\vdots$	$\vdots$	$\vdots$		
s^0	a_0			

Row Computation Formulas

$$b_1 = \frac{a_{n-1} \cdot a_{n-2} - a_n \cdot a_{n-3}}{a_{n-1}}, \quad b_2 = \frac{a_{n-1} \cdot a_{n-4} - a_n \cdot a_{n-5}}{a_{n-1}}, \quad \dots$$

$$c_1 = \frac{b_1 \cdot a_{n-3} - a_{n-1} \cdot b_2}{b_1}, \quad c_2 = \frac{b_1 \cdot a_{n-5} - a_{n-1} \cdot b_3}{b_1}, \quad \dots$$

Continue until all rows are complete. **Stability:** No sign changes in column 1.

2.2 Special Cases

Special Case	Handling
Zero in first column	Replace zero with small $\varepsilon > 0$, complete the array, then let $\varepsilon \rightarrow 0^+$. Count sign changes.
Entire row of zeros	System has symmetric roots (pairs on imaginary axis or real axis). Form auxiliary polynomial from the row above, differentiate it, and use its coefficients to replace the zero row.

2.3 Worked Example

Example — 3rd Order System

Characteristic equation: $s^3 + 2s^2 + 4s + 8 = 0$

Routh Array:

$$\begin{array}{c|cc}
 s^3 & 1 & 4 \\
 s^2 & 2 & 8 \\
 s^1 & b_1 = \frac{2 \times 4 - 1 \times 8}{2} = \frac{0}{2} = 0 & 0 \\
 s^0 & 8 &
 \end{array}$$

$b_1 = 0 \rightarrow$ zero in first column $\rightarrow$ replace with ε , let $\varepsilon \rightarrow 0^+$.

First column: 1, 2, ε , 8 — as $\varepsilon \rightarrow 0^+$, no sign change.

Result: Marginally stable (one pair of roots on imaginary axis at $s = \pm j2$).

Exam Tips — Routh Criterion

- **Necessary condition first:** All coefficients positive and non-zero — check before building the array.
- **Number of sign changes** in column 1 = number of unstable poles.
- **Marginal stability:** Entire row of zeros $\rightarrow$ auxiliary polynomial roots on $j\omega$ -axis.
- **Range of K for stability:** Set up array with K , find condition on first column > 0 .
- Routh criterion gives *no information* about the actual pole locations — only their count in RHP.

3. Steady-State Error (SSE)

Even a stable system may not track the reference input perfectly in steady state. The **steady-state error** quantifies this residual deviation after all transients have died out.

Definition: The **steady-state error** e_{ss} is the difference between the desired reference input and the actual output as $t \rightarrow \infty$:

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s)$$

where $E(s) = \frac{R(s)}{1 + G(s)H(s)}$ for a unity-feedback system.

3.1 System Type Number

The **type number** of a system is the number of open-loop poles at the origin (i.e., the number of pure integrators in $G(s)H(s)$).

$$G(s)H(s) = \frac{K(1 + T_1s)(1 + T_2s) \cdots}{s^N(1 + T_as)(1 + T_bs) \cdots}$$

Here N is the **type number** (0, 1, 2, ...). More integrators $\rightarrow$ better steady-state tracking.

3.2 Error Constants and SSE Formulas

Input	Type 0	Type 1	Type 2
Step $R(s) = \frac{1}{s}$	$\frac{1}{1 + K_p}$	0	0
Ramp $R(s) = \frac{1}{s^2}$	∞	$\frac{1}{K_v}$	0
Parabola $R(s) = \frac{1}{s^3}$	∞	∞	$\frac{1}{K_a}$

Error Constants

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) \quad (\text{Position constant})$$

$$K_v = \lim_{s \rightarrow 0} s G(s)H(s) \quad (\text{Velocity constant})$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) \quad (\text{Acceleration constant})$$

3.3 Steady-State Accuracy

Steady-State Accuracy is the ability of a control system to track the reference input with minimum error in steady state. It is directly related to the **type number** and the **gain** of the open-loop transfer function. Higher type number and higher gain both improve steady-state accuracy.

Exam Tips — SSE

- SSE depends on **system type** and **input type** — memorise the table above.
- Increasing open-loop gain K reduces SSE for the same type, but may reduce stability.
- Adding an integrator (increasing type) eliminates SSE for lower-order inputs but risks instability.
- For **non-unity feedback**, use $E(s) = R(s) - C(s)$ and apply final value theorem carefully.
- **Final Value Theorem** valid only if $sE(s)$ has all poles in LHP (system must be stable).

4. Disturbance Rejection, Sensitivity and Robustness

4.1 Disturbance Rejection

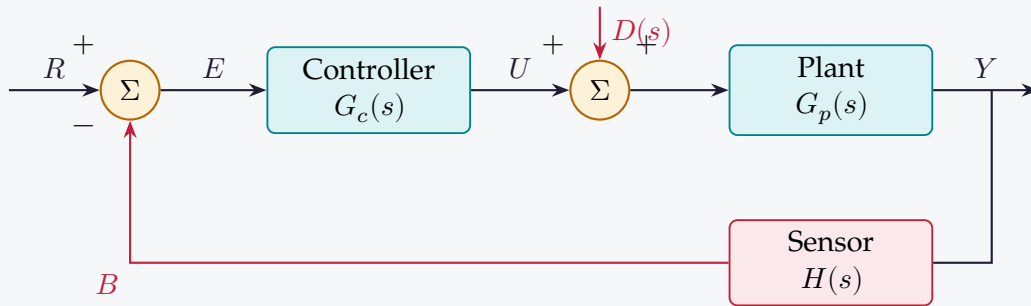
Real systems are subject to unwanted inputs (disturbances) such as load changes, noise, and environmental variations. A good control system must **reject disturbances** — i.e., minimise their effect on the output.

Disturbance Rejection: The ability of a closed-loop system to suppress the effect of disturbance inputs $D(s)$ on the output $Y(s)$. The **disturbance transfer function** is:

$$\left. \frac{Y(s)}{D(s)} \right|_{R=0} = \frac{G_p(s)}{1 + G_c(s)G_p(s)H(s)}$$

High controller gain $G_c(s)$ reduces this ratio, improving disturbance rejection.

Closed Loop with Disturbance



4.2 Sensitivity

Sensitivity Function $S(s)$: Measures how much the closed-loop transfer function $T(s)$ changes relative to changes in the plant $G(s)$:

$$S(s) = \frac{1}{1 + G(s)H(s)}$$

Complementary Sensitivity $T(s)$ $T(s) = 1 - S(s) = \frac{G(s)H(s)}{1 + G(s)H(s)}$

Note: $S(s) + T(s) = 1$ always. High loop gain $\Rightarrow S \approx 0$ (low sensitivity).

4.3 Robustness

Robustness: The ability of a control system to maintain acceptable performance despite *uncertainties* in the plant model (parameter variations, unmodelled dynamics, external disturbances). A robust system has:

- Low sensitivity to plant parameter changes
- Good disturbance rejection
- Adequate gain and phase margins

Property	Open Loop	Closed Loop
Sensitivity to ΔG	High — directly affects output	Reduced by factor $1 + GH$
Disturbance rejection	None	Good (high gain helps)
Robustness	Poor	Better (with proper design)
Bandwidth	Narrow	Wider

5. PID Controllers

A **PID controller** combines three control actions — Proportional, Integral, and Derivative — to achieve fast response, zero steady-state error, and good damping simultaneously.

PID Control Law (Time Domain)

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Transfer Function (s-domain):

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}$$

5.1 Proportional (P) Controller

P Controller: $u(t) = K_p e(t)$, $G_c(s) = K_p$

Output is proportional to the current error. Increasing K_p reduces SSE but increases overshoot and may cause instability.

Effect of increasing K_p	Result
Rise time	Decreases
Overshoot	Increases
Settling time	Small change
Steady-state error	Decreases (but never zero for Type 0)
Stability	Degrades at very high K_p

5.2 Integral (I) Controller

I Controller: $u(t) = K_i \int e(t) dt$, $G_c(s) = K_i/s$

Integrates the error over time. Eliminates steady-state error completely (adds a pole at origin, increasing system type by 1). However, it slows the response and can cause

instability (integral windup).

5.3 Derivative (D) Controller

D Controller: $u(t) = K_d \frac{de}{dt}$, $G_c(s) = K_d s$

Responds to the *rate of change* of error — anticipates future error. Improves damping and reduces overshoot. **Never used alone** (amplifies noise; no steady-state correction). Always combined with P or PI.

5.4 Combined PID — Effect Summary

Controller	Rise Time	Overshoot	Settling	SSE
P	Decrease	Increase	Small Δ	Decrease
I	Decrease	Increase	Increase	Eliminate
D	Small Δ	Decrease	Decrease	No effect
PI	Decrease	Increase	Increase	Eliminate
PD	Decrease	Decrease	Decrease	No effect
PID	Decrease	Decrease	Decrease	Eliminate

Exam Tips — PID

- **P alone:** Always has residual SSE for Type 0 systems.
- **I action:** Eliminates SSE but causes *integral windup* — output saturates if error persists.
- **D action:** Improves transient response; **never used alone** — amplifies high-frequency noise.
- **PID transfer function** has a zero at origin (from K_i/s) and two zeros from numerator.
- **Ziegler–Nichols method** is the standard tuning technique for PID (may appear in viva).

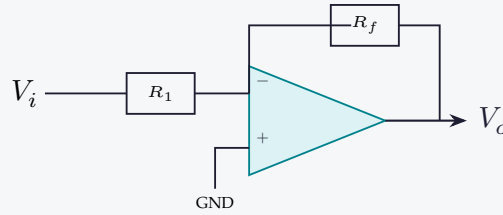
6. Realization of PID Controllers

6.1 Op-Amp Realization

Op-amp circuits implement P, I, D, and PID actions using resistors and capacitors. The inverting op-amp configuration is standard.

Proportional Controller

P Controller — Inverting Amplifier



$$G_c(s) = -\frac{R_f}{R_1} = -K_p \quad (K_p = R_f/R_1)$$

PI Controller

Replace R_f with a series combination of R_f and capacitor C_f :

PI Controller — Op-Amp Transfer Function

$$G_c(s) = -\frac{Z_f}{Z_1} = -\frac{R_f + 1/(sC_f)}{R_1} = -\left(\frac{R_f}{R_1} + \frac{1}{sR_1C_f}\right) = -\left(K_p + \frac{K_i}{s}\right)$$

$$K_p = \frac{R_f}{R_1}, \quad K_i = \frac{1}{R_1C_f}$$

PD Controller

Replace input R_1 with a series combination of R_1 and capacitor C_1 :

PD Controller — Op-Amp Transfer Function

$$G_c(s) = -\frac{R_f}{R_1 + 1/(sC_1)} = -\frac{sR_fC_1}{1 + sR_1C_1} \approx -(K_p + K_d s) \text{ for small } R_1$$

$$K_p = \frac{R_f}{R_1}, \quad K_d = R_fC_1$$

PID Controller (Op-Amp)

PID Controller — Op-Amp Transfer Function

Use $Z_1 = R_1 + 1/(sC_1)$ as input impedance and $Z_f = R_f \parallel (1/sC_f)$ as feedback impedance:

$$G_c(s) = -\frac{Z_f}{Z_1} = -\frac{R_f/(1 + sR_fC_f)}{R_1 + 1/(sC_1)}$$

Simplified ideal form:

$$G_c(s) = - \left(K_p + \frac{K_i}{s} + K_d s \right)$$

$$K_p = \frac{R_f}{R_1}, \quad K_i = \frac{1}{R_1 C_f}, \quad K_d = R_f C_1$$

The negative sign is corrected by cascading an inverting unity-gain stage.

6.2 Digital Implementation of PID

In digital (discrete-time) systems, the PID algorithm is implemented in software using numerical approximations of integration and differentiation.

Discrete PID Algorithm: With sampling period T_s , the continuous PID is discretised as:

- **Proportional:** $P[k] = K_p e[k]$
- **Integral** (rectangular/trapezoidal): $I[k] = I[k-1] + K_i T_s e[k]$
- **Derivative** (backward difference): $D[k] = K_d \frac{e[k] - e[k-1]}{T_s}$
- **Output:** $u[k] = P[k] + I[k] + D[k]$

Pulse Transfer Function (Z-domain)

Using backward Euler ($s \approx \frac{z-1}{T_s z}$):

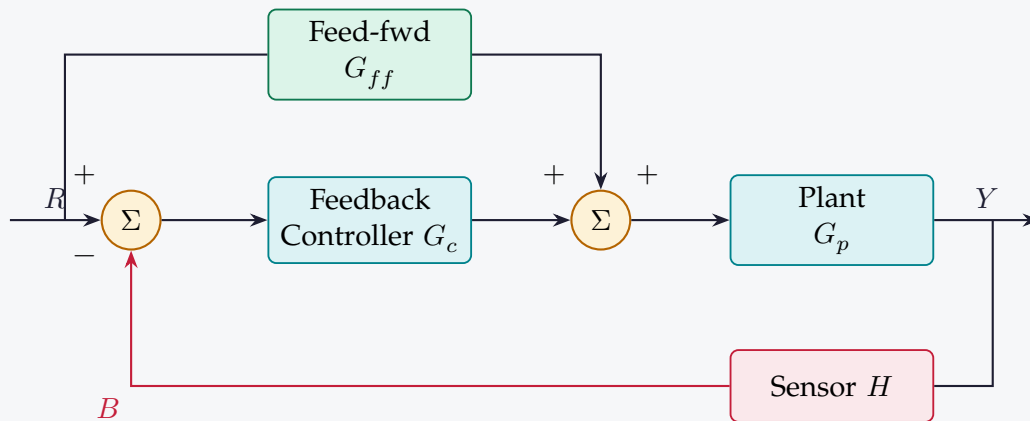
$$G_c(z) = K_p + K_i \frac{T_s z}{z-1} + K_d \frac{z-1}{T_s z}$$

Aspect	Analog (Op-Amp)	Digital (Microcontroller)
Implementation	Hardware (R, C, op-amp)	Software (algorithm in MCU/DSP)
Tuning	Change component values	Change K_p, K_i, K_d in code
Noise	Susceptible to component drift	Quantisation noise; sampling delay
Flexibility	Fixed once built	Easily reprogrammable
Cost	Low for simple systems	Higher (processor needed)

7. Feed-Forward Control

Feed-Forward Control: A control strategy in which the controller acts on a *measured disturbance* or the *reference input directly*, before the disturbance affects the output. It does **not** use feedback — it anticipates and cancels the disturbance proactively.

Feed-Forward + Feedback Configuration



Ideal Feed-Forward Condition

For perfect disturbance cancellation, the feed-forward controller should satisfy:

$$G_{ff}(s) = -\frac{G_d(s)}{G_p(s)}$$

where $G_d(s)$ is the disturbance transfer function and $G_p(s)$ is the plant. In practice, exact cancellation is impossible due to model uncertainty — hence feedback is always retained alongside feed-forward.

Aspect	Feedback Only	Feed-Forward + Feedback
Disturbance handling	Reacts after output is affected	Anticipates and cancels disturbance
Model dependency	Low (error-driven)	High (needs accurate plant model)
Stability risk	Inherent (loop gain)	Lower (FF is open-loop)
Typical use	General purpose	Known, measurable disturbances

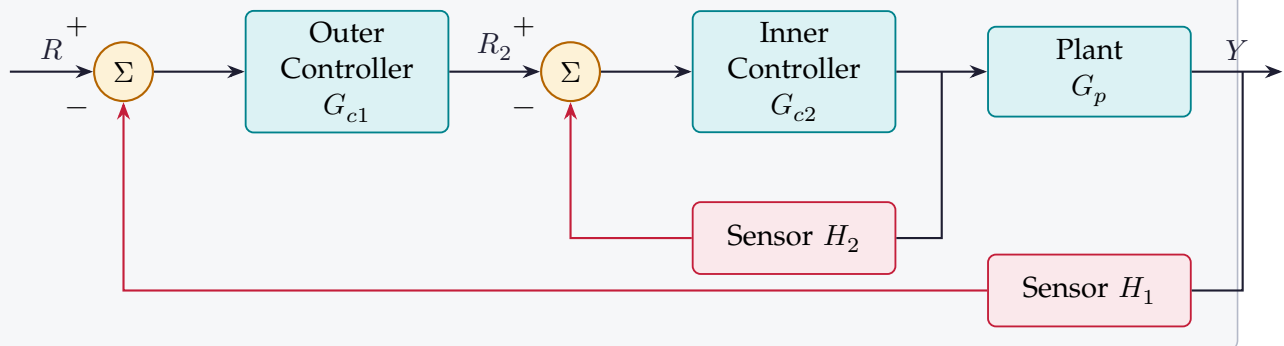
8. Multi-Loop Control Configurations

Multi-Loop Control: Control architectures that use more than one feedback loop to improve performance. The most common configurations are **cascade (series)** control and **ratio control**.

8.1 Cascade (Inner–Outer Loop) Control

Cascade Control: Two feedback loops — an **inner (secondary) loop** and an **outer (primary) loop**. The output of the outer controller becomes the *set-point* for the inner controller. The inner loop handles fast disturbances; the outer loop handles the primary variable.

Cascade Control Block Diagram



Advantages of Cascade Control:

- Inner loop rejects disturbances *before* they propagate to the outer loop.
- Faster overall response — inner loop is tuned for speed.
- Outer loop can be tuned independently for accuracy.

8.2 Ratio Control

Ratio Control: Maintains a fixed ratio between two process variables (e.g., fuel-to-air ratio in a combustion system). One variable is the *wild* (uncontrolled) stream; the other is the *controlled* stream adjusted to maintain the ratio.

$$\text{Set-point of controlled stream} = K_r \times (\text{measured wild stream})$$

Quick Revision — Module 2

Feedback Control Systems | EC601

Key Formulas at a Glance

BIBO Stability	All closed-loop poles in LHP ($\text{Re}(s) < 0$)
Characteristic Equation	$1 + G(s)H(s) = 0$
Sensitivity Function	$S(s) = \frac{1}{1 + G(s)H(s)}$
Disturbance TF	$\frac{Y}{D}\bigg _{R=0} = \frac{G_p}{1 + G_c G_p H}$
SSE (step, Type 0)	$e_{ss} = \frac{1}{1 + K_p}$
SSE (ramp, Type 1)	$e_{ss} = \frac{1}{K_v}$
Position constant	$K_p = \lim_{s \rightarrow 0} G(s)H(s)$
Velocity constant	$K_v = \lim_{s \rightarrow 0} sG(s)H(s)$
PID (s-domain)	$G_c(s) = K_p + K_i/s + K_d s$
PI op-amp	$K_p = R_f/R_1, \quad K_i = 1/(R_1 C_f)$
PID op-amp	$K_p = R_f/R_1, \quad K_i = 1/(R_1 C_f), \quad K_d = R_f C_1$
Digital PID integral	$I[k] = I[k-1] + K_i T_s e[k]$
Digital PID derivative	$D[k] = K_d(e[k] - e[k-1])/T_s$
Ideal FF condition	$G_{ff} = -G_d/G_p$

One-Line Definitions

- **BIBO Stability:** Bounded input $\Rightarrow$ bounded output; all CLTF poles in LHP.
- **Relative Stability:** How far poles are from imaginary axis; measured by gain/phase margin.
- **Routh Criterion:** Algebraic test for stability — no sign changes in first column of Routh array.
- **SSE:** Residual error as $t \rightarrow \infty$; depends on system type and input type.
- **System Type:** Number of open-loop poles at origin (integrators in $G(s)H(s)$).
- **Sensitivity:** $S = 1/(1 + GH)$; measures effect of plant variation on closed-loop TF.
- **Robustness:** Ability to maintain performance despite model uncertainty and disturbances.
- **P Controller:** Proportional to error; reduces SSE but cannot eliminate it (Type 0).

- **I Controller:** Integrates error; eliminates SSE; risks integral windup and instability.
- **D Controller:** Responds to rate of change; improves damping; never used alone.
- **PID:** Combines all three — fast response, zero SSE, good damping.
- **Feed-Forward:** Acts on disturbance before it affects output; requires accurate plant model.
- **Cascade Control:** Inner loop handles fast disturbances; outer loop controls primary variable.

Frequently Asked / Viva Questions

Q1. What is the necessary condition for stability using Routh?

All coefficients of the characteristic polynomial must be positive and non-zero.

Q2. What does a sign change in Routh's first column indicate?

Each sign change corresponds to one closed-loop pole in the right half s -plane.

Q3. What is the difference between absolute and relative stability?

Absolute: stable or not. Relative: how stable (distance of poles from imaginary axis).

Q4. Why does adding an integrator eliminate SSE?

It increases system type by 1, making the velocity/position constant infinite for the corresponding input.

Q5. Why is D action never used alone?

It has no DC gain ($G_c(0) = 0$), cannot correct steady-state error, and amplifies high-frequency noise.

Q6. What is integral windup?

When the integrator accumulates a large value during saturation, causing a large overshoot when the system comes out of saturation.

Q7. What is the sensitivity function?

$S(s) = 1/(1 + GH)$; measures how closed-loop TF changes with plant variation. $S + T = 1$.

Q8. What is the advantage of cascade control?

Inner loop rejects disturbances faster before they affect the outer (primary) loop.

Q9. What is the ideal feed-forward condition?

$G_{ff} = -G_d/G_p$ — cancels disturbance perfectly (requires exact plant model).

Q10. How is PID implemented digitally?

Using backward Euler or Tustin (bilinear) approximation; integral is a running sum, derivative is a backward difference.

SSE Quick Reference Table

Input	Type 0	Type 1	Type 2
Step	$\frac{1}{1 + K_p}$	0	0
Ramp	∞	$\frac{1}{K_v}$	0
Parabola	∞	∞	$\frac{1}{K_a}$